Enlargements

AN ENLARGEMENT IS A TRANSFORMATION OF AN OBJECT THAT PRODUCES AN IMAGE OF THE SAME SHAPE BUT OF DIFFERENT SIZE.

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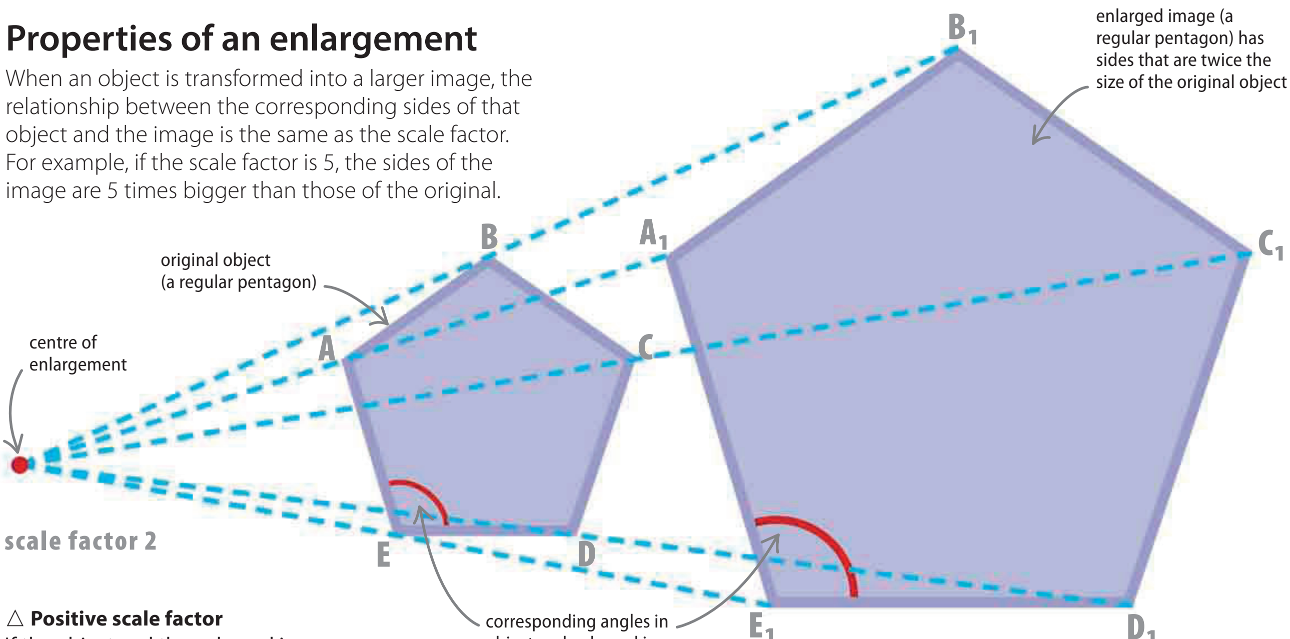
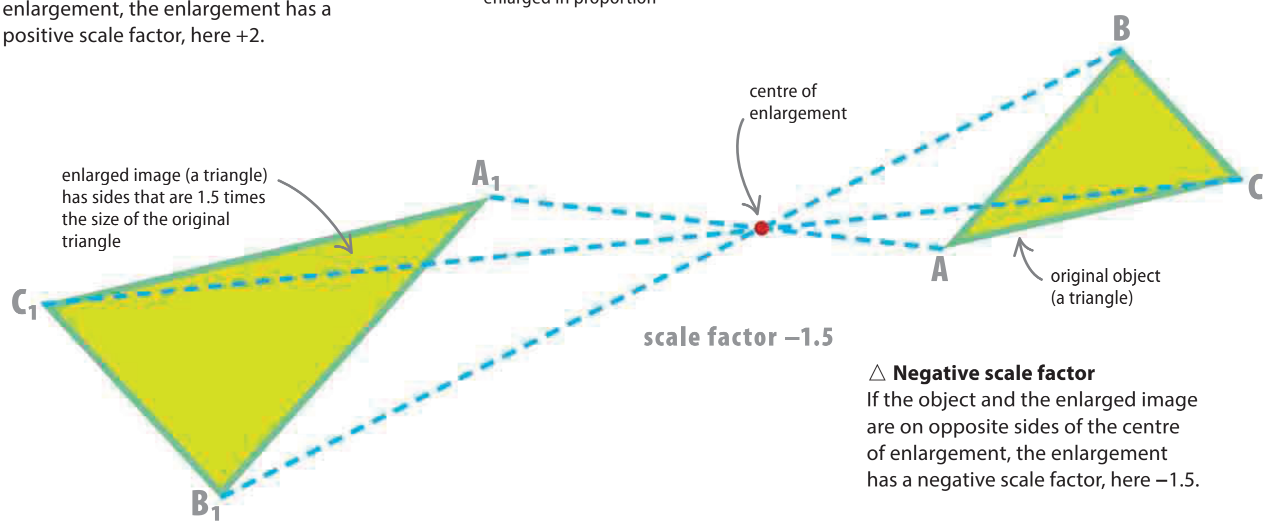
Enlargements are constructed through a fixed point known as the centre of enlargement. The image can be larger or smaller. The change in size is determined by a number called the scale factor.

Properties of an enlargement

When an object is transformed into a larger image, the relationship between the corresponding sides of that object and the image is the same as the scale factor. For example, if the scale factor is 5, the sides of the image are 5 times bigger than those of the original.



△ **Positive scale factor**
If the object and the enlarged image are on the same side of the centre of enlargement, the enlargement has a positive scale factor, here +2.



△ **Negative scale factor**
If the object and the enlarged image are on opposite sides of the centre of enlargement, the enlargement has a negative scale factor, here –1.5.